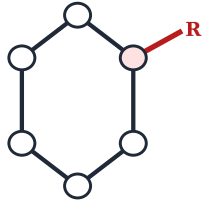
1. Input Molecule

$G = (V, E)$



SMILES Featurization
Node $\mathbf{x}_i$ / Bond $\mathbf{e}_{ij}$

2. GNN Prediction

Baseline Inference

GINE Encoder

$\mathbf{h}_G = \sum a_i \mathbf{h}_i$

$\hat{y}_{\text{orig}} \in [0, 1]$

Computes initial
toxicity probability

3. Substructure Attribution

SMARTS Alert

Attention Filter:

$a_i > \tau_{\text{cutoff}}$

Matched Subgraph $\mathcal{S}$

Identifies attributed
toxicophore candidate

4. In-Silico Deletion

Counterfactual Gen

Delete $\mathcal{S}$ from G:

$G' = G \setminus \mathcal{S}$

RDKit Sanitization

Removes invalid valencies;
retains 68.1% valid set

5. Re-Prediction

Evaluate G'

GNN on G':

$\hat{y}_{\text{cf}} = f_\theta(G')$

$\Delta y = \hat{y}_{\text{orig}} - \hat{y}_{\text{cf}}$

Measures causal shift
upon toxicophore deletion

YES

$\Delta y \geq \delta$**?**

$(\delta = 0.1)$

NO